

Bulan 1: Python Fundamentals & Basic Programming

Minggu 1: Getting Started with Python

Hari 1: Setup & Hello World

Topik: Installation, IDE setup, print(), variables, comments

Proyek 1: GreetMaster

Program salam interaktif yang menyapa user dengan nama dan waktu

- Input nama user
- Deteksi waktu (pagi/siang/malam)
- Format output dengan string manipulation

Proyek 2: CalcuQuick

Kalkulator sederhana 2 angka

- Input 2 angka
- Operasi +, -, *, /
- Print hasil dengan format rapi

Hari 2: Data Types & Operators

Topik: int, float, string, boolean, arithmetic operators, comparison

Proyek 1: TempConverter

Konverter suhu multi-format

- Celsius ke Fahrenheit & Kelvin
- Input validation (harus angka)
- Format output 2 desimal

Proyek 2: AgeCalculator

Hitung umur dalam berbagai satuan

- Input tahun lahir
- Hitung umur dalam tahun, bulan, hari, jam
- Teknologi: datetime, arithmetic operations

Hari 3: Strings & String Methods

Topik: String indexing, slicing, methods (upper, lower, strip, replace, split)

Proyek 1: TextCleaner

Pembersih dan formatter teks

- Remove whitespace berlebih
- Capitalize setiap kata
- Count karakter, kata, kalimat

Proyek 2: PasswordChecker

Validator password sederhana

- Cek panjang minimal 8 karakter
- Cek ada uppercase, lowercase, angka
- Return strength level (weak/medium/strong)

Hari 4: Lists & List Methods

Topik: Creating lists, indexing, slicing, append, extend, insert, remove, sort

Proyek 1: ShoppingCart

Keranjang belanja interaktif

- Tambah/hapus item
- Tampilkan semua item
- Hitung total item
- Sort alphabetically

Proyek 2: GradeBook

Pembukuan nilai siswa

- Input multiple nilai
- Hitung rata-rata
- Find nilai tertinggi/terendah
- Sort nilai

Hari 5: Dictionaries

Topik: Key-value pairs, accessing, adding, updating, methods (keys, values, items)

Proyek 1: ContactHub

Buku kontak sederhana

- Simpan nama & nomor telepon
- Cari kontak by nama
- Update nomor
- List semua kontak

Proyek 2: InventoryLite

Inventory counter

- Track item & quantity
- Add/update stock
- Low stock alert (< 5)
- Display total items

Hari 6: Tuples & Sets

Topik: Tuples (immutable), sets (unique values), operations

Proyek 1: CoordinateMapper

Penyimpan koordinat lokasi

- Store coordinates as tuples
- Calculate distance between points
- Teknologi: tuples, math operations

Proyek 2: TagManager

Manager tag unik

- Input multiple tags
- Remove duplicates dengan sets
- Find common tags antara 2 lists
- Union & intersection operations

Hari 7: Review & Mini Project

Mini Project: BudgetTracker

Pelacak keuangan sederhana

- Input pemasukan & pengeluaran (dict)

- Kategorikan transaksi (list)
- Hitung saldo
- Show transaksi by kategori
- Find pengeluaran terbesar
- Teknologi: dict, list, string formatting, basic calculations

Minggu 2: Control Flow & Functions

Hari 8: If-Else Statements

Topik: if, elif, else, nested conditionals, logical operators (and, or, not)

Proyek 1: BMICalculator

Kalkulator BMI dengan kategori

- Input tinggi & berat
- Hitung BMI
- Kategorikan (underweight, normal, overweight, obese)
- Berikan rekomendasi

Proyek 2: QuizMaster

Quiz 5 pertanyaan

- Multiple choice questions
- Track score
- Conditional feedback per jawaban
- Final grade dengan kondisi

Hari 9: For Loops

Topik: for loops, range(), iterating lists/dicts, enumerate()

Proyek 1: MultiplicationTable

Generator tabel perkalian

- Input angka 1-10
- Print tabel perkalian 1-12
- Format output rapi (aligned)

Proyek 2: WordCounter

Analisis teks

- Input kalimat panjang
- Count frequency setiap kata
- Display top 5 kata terbanyak
- Teknologi: for loop, dict, sorting

Hari 10: While Loops

Topik: while loops, break, continue, infinite loops prevention

Proyek 1: GuessTheNumber

Game tebak angka

- Random number 1-100
- User tebak dengan hints (higher/lower)
- Count attempts
- Option play again
- Teknologi: while, random module

Proyek 2: ATMSimulator

Simulasi ATM sederhana

- Menu loop (check balance, deposit, withdraw, exit)
- Validation (cukup saldo atau tidak)
- Transaction history
- While loop untuk menu

Hari 11: Functions Basics

Topik: def, parameters, return, docstrings

Proyek 1: MathLibrary

Library fungsi matematika

- Function: is_prime(n)
- Function: factorial(n)
- Function: fibonacci(n)
- Function: gcd(a, b)

- Test semua functions

Proyek 2: StringUtils

Utility functions untuk string

- Function: reverse_string()
- Function: is_palindrome()
- Function: count_vowels()
- Function: remove_duplicates()

Hari 12: Function Parameters

Topik: Default parameters, *args, **kwargs, keyword arguments

Proyek 1: OrderSystem

Sistem pemesanan fleksibel

- Function dengan default tax & discount
- Handle multiple items (*args)
- Handle options (**kwargs: delivery, gift_wrap)
- Calculate total dengan berbagai kombinasi

Proyek 2: ReportGenerator

Generator laporan fleksibel

- Function terima variable number of data
- Optional parameters (format, detailed)
- Return formatted report
- Teknologi: *args, **kwargs, string formatting

Hari 13: Scope & Lambda

Topik: Global vs local scope, lambda functions, map, filter

Proyek 1: DataFilter

Filter & transform data

- List of numbers
- Filter even/odd dengan filter()
- Square numbers dengan map()
- Custom filters dengan lambda

Proyek 2: PriceCalculator

Kalkulator harga dengan transforms

- List of prices
- Apply discount dengan map()
- Filter by price range
- Sort by custom criteria dengan lambda

Hari 14: Review & Mini Project

Mini Project: TaskManagerCLI

Task manager command line

- Functions: add_task, remove_task, mark_complete
- List tasks dengan filter (all, pending, completed)
- Priority system (high, medium, low)
- Due date tracking
- Search tasks
- Statistics (total, completed percentage)
- Teknologi: functions, lists, dicts, loops, conditionals

Minggu 3: File Handling & Error Management

Hari 15: Reading Files

Topik: open(), read(), readline(), readlines(), with statement

Proyek 1: LogAnalyzer

Analisis file log

- Read log file line by line
- Count error vs success entries
- Extract timestamps
- Summary report

Proyek 2: ConfigReader

Pembaca konfigurasi

- Read config file (key=value format)

- Parse ke dictionary
- Validate required keys
- Display configurations

Hari 16: Writing Files

Topik: write(), writelines(), append mode, file modes

Proyek 1: JournalApp

Aplikasi jurnal harian

- Write daily entries dengan timestamp
- Append mode untuk entries baru
- Read past entries
- Search by date

Proyek 2: ReportExporter

Export data ke file

- Generate sales report
- Write to text file formatted
- Append summary statistics
- Teknologi: file writing, string formatting, datetime

Hari 17: CSV Files

Topik: csv module, reading CSV, writing CSV, DictReader, DictWriter

Proyek 1: StudentGradeManager

Manager nilai dalam CSV

- Read student data dari CSV
- Calculate averages
- Add new students
- Export results to new CSV

Proyek 2: ExpenseTracker

Track pengeluaran di CSV

- Log expenses (date, category, amount, description)
- Read & analyze expenses

- Calculate by category
- Monthly summary

Hari 18: JSON Files

Topik: json module, json.load(), json.dump(), working with JSON data

Proyek 1: UserProfileManager

Manager profil user JSON

- Save user profiles (name, age, preferences)
- Load profiles
- Update specific fields
- Pretty print JSON

Proyek 2: APIResponseParser

Parser simulasi API response

- Read JSON API response (sample data)
- Extract specific fields
- Transform & save to new JSON
- Handle nested JSON objects

Hari 19: Exception Handling

Topik: try-except, multiple exceptions, finally, raise

Proyek 1: SafeCalculator

Kalkulator dengan error handling

- Handle division by zero
- Handle invalid input (non-numeric)
- Handle undefined operations
- User-friendly error messages

Proyek 2: FileValidator

Validator file dengan error handling

- Try open file
- Handle FileNotFoundError
- Handle PermissionError

- Validate file format
- Log errors

Hari 20: Custom Exceptions

Topik: Creating custom exceptions, exception hierarchy

Proyek 1: BankingSystem

Sistem banking dengan custom errors

- InsufficientFundsError
- InvalidAmountError
- AccountNotFoundError
- Handle transactions safely

Proyek 2: FormValidator

Validator form dengan custom exceptions

- EmailValidationError
- PhoneValidationError
- AgeValidationError
- Validate & report specific errors

Hari 21: Review & Mini Project

Mini Project: ContactManagerPro

Contact manager lengkap dengan persistence

- Save/load contacts dari JSON
- Add, edit, delete contacts
- Search & filter contacts
- Export to CSV
- Import from CSV
- Error handling untuk semua operations
- Backup functionality
- Statistics dashboard
- Teknologi: JSON, CSV, exceptions, functions, file I/O

Minggu 4: Object-Oriented Programming (OOP)

Hari 22: Classes & Objects Basics

Topik: class, **init**, self, attributes, methods

Proyek 1: BankAccount

Class untuk bank account

- Attributes: account_number, holder_name, balance
- Methods: deposit(), withdraw(), get_balance()
- Create multiple account objects

Proyek 2: Book

Class untuk buku

- Attributes: title, author, pages, price
- Methods: get_info(), apply_discount()
- Create library of books

Hari 23: Encapsulation

Topik: Private attributes (**__**), getters, setters, **@property**

Proyek 1: Employee

Class employee dengan private salary

- Private **__salary** attribute
- Getter & setter methods
- Validation (salary > 0)
- Calculate annual salary

Proyek 2: Product

Class produk e-commerce

- Private **__price**
- Property untuk discount price
- Stock management (private **__stock**)
- Validation rules

Hari 24: Inheritance

Topik: Parent & child classes, super(), method overriding

Proyek 1: VehicleSystem

Hierarki kendaraan

- Parent: Vehicle (brand, model, year)
- Child: Car (doors, trunk_size)
- Child: Motorcycle (type, cc)
- Override methods

Proyek 2: EmployeeHierarchy

Sistem karyawan

- Parent: Employee (name, id, base_salary)
- Child: Manager (bonus, team_size)
- Child: Developer (programming_languages, projects)
- Calculate different salary formulas

Hari 25: Polymorphism

Topik: Method overriding, duck typing, isinstance()

Proyek 1: ShapeCalculator

Hitung area berbagai bentuk

- Parent: Shape
- Children: Circle, Rectangle, Triangle
- Each with area() & perimeter() methods
- Calculate total area of multiple shapes

Proyek 2: PaymentProcessor

Sistem pembayaran

- Parent: Payment
- Children: CreditCard, DebitCard, EWallet
- Each with process_payment() method
- Different validation & fees

Hari 26: Special Methods

Topik: **str**, **repr**, **len**, **eq**, operator overloading

Proyek 1: Vector

Class vector matematika

- **init** dengan x, y coordinates
- **add** untuk penjumlahan vector
- **str** untuk display
- **eq** untuk perbandingan

Proyek 2: Money

Class untuk uang dengan currency

- **init** dengan amount & currency
- **add**, **sub** (validasi currency sama)
- **str** formatted (Rp 100,000)
- **lt**, **gt** untuk comparison

Hari 27: Class Methods & Static Methods

Topik: **@classmethod**, **@staticmethod**, class variables

Proyek 1: DateUtility

Utility class untuk dates

- Static methods: **is_leap_year()**, **days_in_month()**
- Class method: **from_string()** parser
- Class variable: **supported_formats**

Proyek 2: DatabaseConnection

Singleton pattern

- Class variable: **__instance**
- Class method: **get_instance()**
- Static methods: **validate_connection()**
- Ensure single database connection

Hari 28: Review & Mini Project

Mini Project: LibraryManagementSystem

Sistem perpustakaan OOP lengkap

Classes:

- Book (title, author, isbn, available)
- Member (name, member_id, borrowed_books)
- Library (books collection, members collection)

Features:

- Add/remove books
- Register members
- Borrow/return system (max 3 books per member)
- Search books (by title, author, isbn)
- Fine calculation (late returns)
- Statistics (most borrowed, active members)
- Save/load dari JSON
- Teknologi: OOP (inheritance, encapsulation), JSON, datetime, exceptions

Minggu 5: Advanced Python Concepts

Hari 29: List Comprehensions

Topik: List comprehensions, conditional comprehensions, nested

Proyek 1: DataTransformer

Transform data dengan comprehensions

- Square even numbers dari list
- Filter strings by length
- Flatten nested lists
- Matrix operations

Proyek 2: TextAnalyzer

Analisis teks dengan comprehensions

- Extract words dengan length > 5

- Count consonants in each word
- Create dict {word: length}
- Find palindromes

Hari 30: Generators & Iterators

Topik: yield, generator functions, iterators, next()

Proyek 1: FibonacciGenerator

Generate infinite fibonacci

- Generator function untuk fibonacci
- Lazy evaluation
- Get first N fibonacci numbers
- Memory efficient

Proyek 2: LogStreamer

Stream large log files

- Generator untuk read large file line by line
- Filter logs by level
- Memory efficient processing
- Teknologi: generators, file handling

Progress Check Bulan 1: ☒ Python syntax & data types ☒ Control flow (if, loops) ☒ Functions & scope
☒ File I/O (text, CSV, JSON) ☒ Error handling ☒ OOP fundamentals ☒ Advanced concepts
(comprehensions, generators)

Total Proyek: 60+ proyek kecil & 4 mini projects **Siap lanjut:** Bulan 2 - Advanced Python & Web Basics